

# Algorithm for Decomposition Using Functional Dependencies -- DBMS

**Definition:** Decomposition Algorithms split a complex relational schema R into smaller sub-schemas (R1, R2, ..., Rn) to eliminate anomalies while preserving Lossless-Join and Functional Dependencies.

## 1. Detailed Technical Explanation

### 1. Attribute Closure Algorithm (F+)

The closure of an attribute set X under F, denoted  $X^+$ , is the set of all attributes functionally determined by X.

#### Algorithm Steps:

Input:  
Output:  
  
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3. Ret

### 2. Lossless-Join Decomposition Testing

A decomposition of R into R1 and R2 is lossless-join with respect to F if and only if:

(R1 n

\*(Meaning: The common attributes between R1 and R2 must form a Super Key for at least one of the decomposed relations).\*

### 3. 3NF Synthesis Algorithm (Lossless-Join & Dependency Preserving)

Input: Relation R, Set of FDs F

Output: 3NF Decomposition of R

1. Compute Minimal Cover  $F_c$  for F.
2. For each FD  $(X \rightarrow Y)$  in  $F_c$ :  
Create a schema  $R_i = X \cup Y$ .
3. If no schema  $R_i$  contains a candidate key of R:  
Create an additional schema containing any candidate key of R.
4. Eliminate redundant schemas (if  $R_i \subseteq R_j$ , remove  $R_i$ ).

## 2. Core Concepts & Memory Keywords

- \* Minimal Cover (Canonical Cover): A simplified set of FDs with single attributes on the right-hand side and no extraneous attributes.

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## 3. Must-Write Points for Exams

- \* Lossless join ensures that joining decomposed tables produces the exact original dataset without fake data.

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## 4. Quick Recall Flow

Find Minimal Cover  $F_c$  -> Compute Attribute Closures  $X^+$  -> Test Lossless ( $R_1 \cap R_2 \rightarrow R_1$ ) -> Preserve FDs -> 3NF Synthesis